

Q-MODE

From the protein pocket to the pharmacophoric lattice for quantum docking

Project	Multidisciplinary Project — pre-processing pipeline for quantum docking
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Subject	Architecture, scientific rationale, and validation of the pipeline

This document describes the Q-MODE pipeline: how it translates a protein binding pocket into a discrete, numerical representation suitable for a quantum docking model, the biological and computational rationale behind each step, and the validation results on five real pockets.

1. Summary

Q-MODE is a pre-processing pipeline that transforms a protein binding pocket (.pdb file) into a discrete, ordered sequence of pharmacophoric sites, encodable on qubits. The core idea is to abandon continuous 3D atomic coordinates — too detailed and variable to be mapped onto a quantum computer — in favor of a 2D integer lattice in which each node carries an interaction label (hydrophobic, hydrogen-bond donor/acceptor, aromatic, ionizable). This discrete abstraction is the condition that makes the problem encodable with a limited number of qubits.

The project implements the algorithmic phase that connects raw structural data to the quantum encoding, a phase left at a theoretical level in the reference literature on the quantum algorithm for identifying docking sites in interaction space.

2. Biological context

2.1 Residues, backbone, and side chain

A protein is a chain of amino acids. When two amino acids bind they form a peptide bond with the loss of a water molecule: the unit that remains along the chain is called a residue. Each residue has two parts. The backbone (N–C α –C–O skeleton) is identical across all 20 amino acids; the side chain (the “R” group attached to the C α) varies from one amino acid to another and is what gives the residue its chemical identity — it makes leucine hydrophobic, aspartate negatively charged, phenylalanine aromatic.

2.2 Binding pocket and pharmacophores

By folding, the protein forms cavities: the binding pocket is the one into which a ligand (e.g., a drug) fits. The residues that form its walls are the ones relevant for docking, and they may be far apart in sequence but close in space. What governs binding is not individual atoms but pharmacophores: the abstract physico-chemical properties that enable interaction. A residue generates one or more pharmacophores. Q-MODE uses six categories, the alphabet of non-covalent interactions:

Type	Physical interaction	Example
HBondDonor	Donates a hydrogen in an H-bond	–NH, –OH (serine)
HBondAcceptor	Accepts a hydrogen (lone pairs)	carbonyl C=O, N
Hydrophobe	Hydrophobic effect (desolvation)	carbon chains (Leu, Val)
Aromatic	π -stacking between rings	Phe, Trp, Tyr
PosIonizable	Positive charge	Lys, Arg
NegIonizable	Negative charge	Asp, Glu

3. The pipeline, step by step

The chain of translations goes from biological data to qubits through seven stages. Each stage is a module of the `amino_lattice/` package.

Module / Role	Description
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pdb_reader.py <i>Step 1 — PDB Reader</i>	Reads the PDB, groups atoms by residue, and reconstructs the chemical topology with RDKit from the canonical SMILES, overlaying the real crystallographic coordinates. Keeps only the 20 standard amino acids. Residues are ordered by the distance of the C α from the pocket centroid: the most central ones, the most likely ligand contacts, come first.
feature_extraction.py <i>Step 2 — Feature Extraction</i>	Extracts the pharmacophores (type + 3D coordinates + intensity) via RDKit's feature factory. Hydrophobicity is quantified using Crippen's atomic contributions to LogP; hydrogen-bond intensity is computed geometrically (see §4.1).
site_selection.py <i>Step 3 — Site Selection</i>	Each residue is reduced to K representative sites (K chosen per residue). A K-Means on the 3D coordinates aggregates redundant pharmacophores: each cluster $\rightarrow$ one site (centroid, dominant type, summed intensity). Sites are then ordered according to chemical connectivity (BFS on the bond graph) or along the PCA principal axis.
lattice_fitting.py <i>Step 4 — PCA Fitting</i>	The 3D coordinates of the K sites are projected onto a 2D plane (PCA, alternatively MDS) to minimize the number of qubits. Projection quality is measured by Kruskal's stress.
snapping.py <i>Step 5 — Snapping</i>	The continuous 2D coordinates are discretized into integer nodes (i, j). The Hungarian algorithm assigns distinct nodes minimizing total displacement, avoiding collisions.
labeling.py <i>Step 6 — Labeling</i>	The type of each site is converted into a numerical vector (one-hot, index, or embedding), producing the final LabeledSite objects (i, j, type, vector).
qubit_chain.py · quantum_encoding.py <i>Step 7 — Quantum Encoding</i>	The sites of all residues are concatenated into a 1D sequence (residues by distance from the center, sites by topology). A sliding window the size of the ligand generates the segments; each site is reduced to two channels (h, hb) and encoded into binary states (First Encoding, for Grover) and amplitudes (Second Encoding, for the Euclidean distance).

Note on the 2D $\rightarrow$ 1D step: this is not a compression of coordinates, but a serialization (an ordered reading of the sites). In the quantum part, the (i, j) coordinates are not used: what matters is the order of the sites and, for each one, the (h, hb) pair.

4. Physical foundations and corrections in this revision

4.1 Hydrogen-bond intensity: from placeholder to geometry

In the previous version, HB intensity was a pseudo-random value derived from the coordinates — a fake energy. It is now computed physically, between different residues of the pocket, as the product of a distance term and an angular term:

$$intensity = \exp[-(d - d_0)^2 / 2\sigma^2] \times \max(0, -\cos \angle D-H\cdots A)$$

where d is the donor...acceptor distance (optimum $d_0 = 2.9 \text{ \AA}$) and $\angle D-H\cdots A$ is the bond angle (optimum at 180° , i.e. linear). Real values across the pockets range from ≈ 0.05 (exposed group, no partner) to ≈ 6 (strong, linear bond): the hb axis that reaches the qubits is now a physically grounded quantity.

4.2 Repaired pharmacophore extraction

Validation revealed that RDKit's feature factory was disabled by an obsolete API: the pipeline produced only HBond and Hydrophobe, never Aromatic/Pos/NegIonizable. Three fixes: (i) modern ChemicalFeatures API; (ii) family-name mapping (Donor→HBondDonor); (iii) a specificity rule in the cluster voting, so that rare, informative types (aromatic, charged) are not masked by the numerous hydrophobic atoms. Crippen contributions now assign hydrophobic intensity instead of flooding the molecule with sites.

4.3 Quantum encoding: informative strings and valid amplitudes

Checking the strings produced revealed two problems: the First Encoding was dominated by zeros (77% all-zero segments, only 8 distinct patterns) and the Second Encoding produced non-normalized amplitudes. Fixes: (i) ionizable sites now flow into the hb channel with geometric intensity (no dead bits); (ii) the binarization threshold is the median of the active channel, not the mean of the range, which collapsed to zero on asymmetric distributions; (iii) the amplitude formula is corrected into cosine/sine form so that $a^2 + b^2 = 1$.

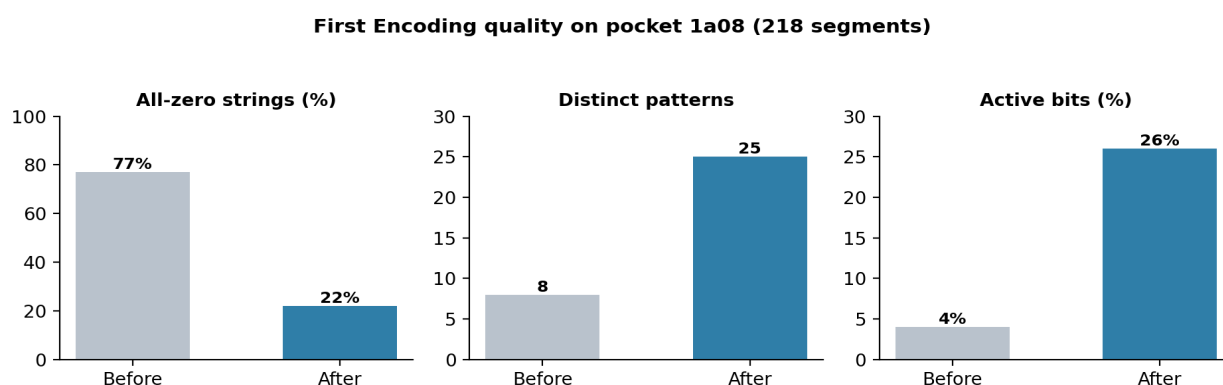


Figure 1 — Effect of the corrections on the First Encoding (pocket 1a08).

5. Validation

Three independent checks (`scripts/validate.py`) verify that the pipeline produces sensible, stable patterns.

5.1 Chemical sanity check on the 20 amino acids

Each standard amino acid is passed through the pipeline, and it is verified that it produces the pharmacophores expected from its known chemistry. Result: **18 / 19** checks passed (aromatic amino acids yield Aromatic, basic ones Poslonizable, acidic ones Neglonizable). Before the corrections, this was 11/19.

5.2 Five real, heterogeneous pockets

The pipeline is applied to five different targets (HIV-1 protease 1a08, trypsin 3ptb, HIV protease 1hsg, dihydrofolate reductase 4dfr, streptavidin 1stp). The pharmacophoric composition is realistic and diverse, and the mean geometric HB intensity falls between 1.8 and 2.3.

Pocket	Residues	Sites	Hyd	Aro	HBD	HBA	Pos	Neg	Mean HB
1a08	38	220	53	3	60	14	47	43	1.79
3ptb	18	97	17	3	27	13	18	19	1.86
1hsg	33	184	56	1	41	12	35	39	2.00
4dfr	40	230	64	8	52	18	46	42	1.93
1stp	17	95	20	7	24	9	17	18	1.92

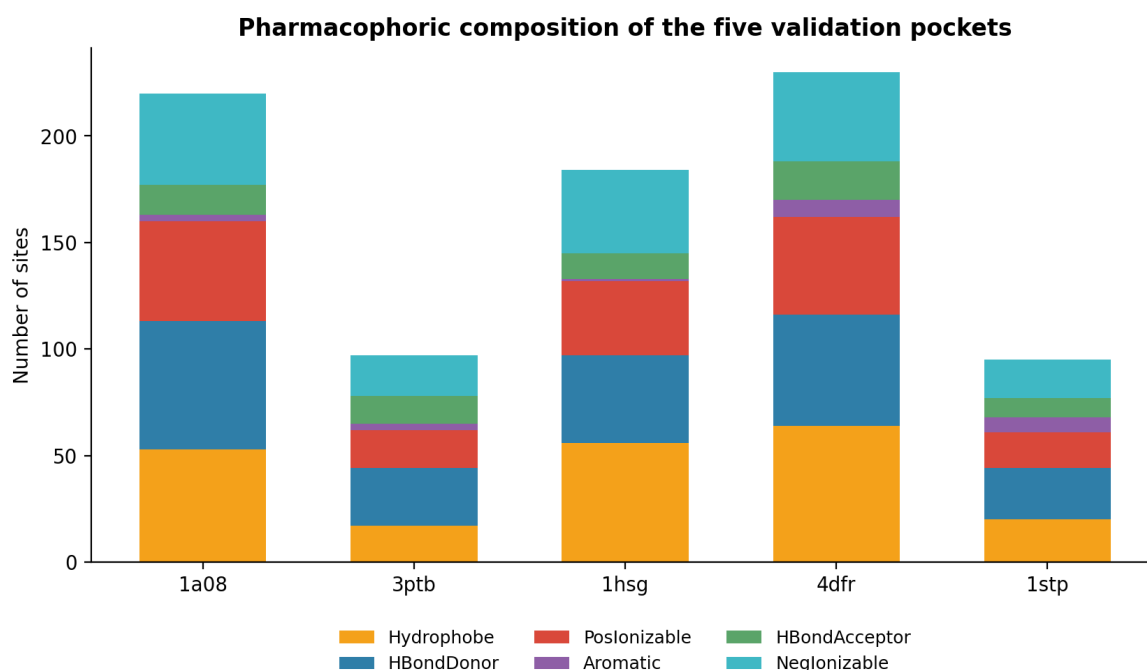


Figure 2 — Pharmacophoric composition of the five validation pockets.

5.3 Determinism

With fixed seeds, two runs on the same pocket produce identical output. The pipeline is reproducible.

6. Limitations and future developments

- **Geometric mapping** — the PDB→RDKit atom coordinate overlay happens by positional order, not chemical identity: for some residues this can assign imperfect coordinates to individual atoms. Downstream stages (clustering, PCA) attenuate this effect.
- **Local lattice** — the (i, j) coordinates are per-residue, not a shared global system: inter-residue geometric relationships are not encoded in the lattice, only in the sequence order.
- **Quantum stage** — the encodings (binary and amplitude, now corrected) are implemented, but not the execution of the quantum algorithms (Grover, swap test): a computed docking score is therefore still missing. This is the natural next step.

7. Conclusions

Q-MODE implements, in a modular and validated way, the entire chain that takes a raw protein pocket to a discrete representation ready for quantum encoding. Compared to the initial version, this revision physically grounds hydrogen-bond intensity, restores the complete extraction of the six pharmacophores, makes the quantum strings informative and normalizes their amplitudes, and adds quantitative validation on five real pockets. The pipeline is solid, reproducible, and ready to be connected to actual quantum algorithms.